

Two Catalogue-Level Alternatives for the Initial GC Birth Radius Scale

Prepared for replacing the spin-based R_e choice in Gao et al. (2023)

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Purpose

Gao et al. (2023) model in-situ globular clusters (GCs) as following a cold-gas-like spatial distribution, then draw their galactocentric birth radii from a spherical deprojected Sérsic profile. The weak step is not the Sérsic sampling itself, but the radius scale:

$$R_e = \frac{\lambda R_{\text{vir}}}{\sqrt{2}}, \quad (1)$$

where λ is a halo spin parameter. This links the compact star-forming gas distribution too directly to the angular momentum of the dark-matter halo. The two methods below keep the downstream Gao-style GC evolution unchanged, but replace this effective radius by a more baryon-motivated estimate.

1 Method 1: Empirical galaxy size–mass–redshift relation

1.1 Physical logic

The formation of in-situ GCs should trace the compact star-forming baryonic component rather than the full dark-matter halo spin. An empirical size–mass–redshift relation therefore provides a cleaner prior for the scale radius of the birth distribution. This method is also compatible with dark-matter-only merger trees, because $M_\star$ can be supplied from the same SMHM relation already used in the GC formation model.

1.2 Concrete formula

For star-forming galaxies at $0 < z \lesssim 3$, a compact and practical form based on van der Wel et al. (2014) is

$$R_{e,\text{SF}} = 8.5 \text{ kpc} \left(\frac{M_\star}{5 \times 10^{10} M_\odot} \right)^{0.22} (1+z)^{-0.75} 10^{\mathcal{N}(0, \sigma_{\log R})}, \quad (2)$$

with

$$\sigma_{\log R} = 0.18 \text{ dex} \quad (3)$$

as a reasonable intrinsic scatter for the late-type/star-forming population.

The coefficient is chosen so that equation (2) reproduces the van der Wel et al. (2014) late-type normalization near the first redshift bin. The slope 0.22 and redshift exponent -0.75 are the measured typical mass and redshift dependences for late-type/star-forming galaxies. This relation is best used above $M_\star \sim 3 \times 10^9 M_\odot$ for $z \lesssim 3$. For lower stellar masses, Nedkova et al. (2021) support a similar

star-forming slope of approximately 0.2 down to $M_\star \sim 10^7 M_\odot$ at $0.2 \leq z \leq 2$, but the extrapolation should be flagged.

A more exact implementation can interpolate the original binned relation:

$$\log_{10} \left(\frac{R_{e,\text{SF}}}{\text{kpc}} \right) = \log_{10} A(z) + \alpha(z) \log_{10} \left(\frac{M_\star}{5 \times 10^{10} M_\odot} \right) + \mathcal{N}(0, \sigma_{\log R}). \quad (4)$$

For late-type/star-forming galaxies, van der Wel et al. (2014) give:

z_{bin}	$\log_{10} A$	α	$\sigma_{\log R}$
0.25	0.86	0.25	0.16
0.75	0.78	0.22	0.16
1.25	0.70	0.22	0.17
1.75	0.65	0.23	0.18
2.25	0.55	0.22	0.19
2.75	0.51	0.18	0.19

Interpolate $\log A$, α , and $\sigma_{\log R}$ in either z or $\log(1+z)$. Interpolating in $\log(1+z)$ is slightly more physical for size evolution.

For $z > 3$, use a high-redshift extension rather than blindly applying the same -0.75 slope. A simple reionization-era calibration based on JWST results is

$$R_{e,\text{SF}} = 0.83 \text{ kpc} \left(\frac{M_\star}{10^{8.5} M_\odot} \right)^{0.25} \left(\frac{1+z}{5} \right)^{-1.24}, \quad 4 \lesssim z \lesssim 8, \quad (5)$$

which reproduces the Miller et al. (2025) statement that the average size at $\log M_\star/M_\odot = 8.5$ grows from about 400 pc at $z = 8$ to about 830 pc at $z = 4$. Treat this as a systematic high- z bracket, not as a universally calibrated relation.

Finally, convert the galaxy star-forming size into a GC birth-radius scale:

$$R_{e,\text{GC}} = f_{\text{GC}} R_{e,\text{SF}}. \quad (6)$$

A clean default is $f_{\text{GC}} = 1$. For systematic tests, use

$$f_{\text{GC}} = 0.3, 1, 3. \quad (7)$$

The compact case is motivated by cluster formation in high-pressure central gas; the extended case brackets merger-induced star formation in clumps, disks, and tidal features.

1.3 Workflow

1. At every GC formation snapshot, obtain z and $M_\star$ for the host galaxy. If the tree is dark-matter-only, infer $M_\star$ from the same SMHM relation already used to set the GC mass budget.
2. Compute $R_{e,\text{SF}}$ from equation (2) or from the table-interpolated equation (4). For $z \gtrsim 4$, compare against equation (5).
3. Apply the GC compactness factor $R_{e,\text{GC}} = f_{\text{GC}} R_{e,\text{SF}}$.
4. Use $R_{e,\text{GC}}$ as the effective radius in the same deprojected Sérsic density profile used by Gao et al. (2023).
5. Draw each GC radius from the cumulative mass profile. Keep $n = 1.0, 1.5, 2.0$ as the main Sérsic-index grid; $n = 1.5$ is a good star-forming-galaxy fiducial.

1.4 Required inputs

- Redshift or scale factor of the GC formation snapshot.
- Stellar mass M_* , from either a hydro counterpart or an SMHM relation.
- Choice of star-forming size relation: equation (2), table interpolation, or high- z JWST bracket.
- Scatter model, usually $\sigma_{\log R} = 0.18\text{--}0.23$ dex.
- GC compactness factor f_{GC} .
- Sérsic index n for spatial sampling.

2 Method 2: Catalogue-level SFR concentration

2.1 Physical logic

This method uses hydro group-catalogue aperture SFRs, not full hydro snapshots. It is suitable when the semi-analytic model is run on a dark-matter-only tree but hydro–DM counterpart catalogues are available. It estimates how concentrated the instantaneous star formation is relative to the stellar half-mass radius, then converts that concentration into an effective radius for the GC birth distribution.

This is closer to Gao et al.’s stated assumption than the spin model: if GCs follow cold star-forming gas, then the relevant radius should respond to where the SFR is concentrated, not to the halo spin alone.

2.2 Required catalogue fields

For a matched hydro subhalo, read:

Quantity	Field	Unit / note
Total SFR	SubhaloSFR	$M_\odot \text{ yr}^{-1}$
SFR inside stellar half-mass radius	SubhaloSFRinHalfRad	$M_\odot \text{ yr}^{-1}$
SFR inside twice stellar half-mass radius	SubhaloSFRinRad	$M_\odot \text{ yr}^{-1}$
Stellar half-mass radius	SubhaloHalfmassRadType[:, 4]	ckpc/ h
Stellar mass	SubhaloMassType[:, 4]	$10^{10} M_\odot / h$
Stellar particle count	SubhaloLenType[:, 4]	resolution cut
Hydro–DM match	matching catalogue	e.g. SubhaloIndexDark_LHaloT

Convert the stellar half-mass radius to physical kpc:

$$R_{\star,1/2}^{\text{phys}}[\text{kpc}] = R_{\star,1/2}^{\text{cat}}[\text{ckpc}/h] \frac{a}{h}. \quad (8)$$

2.3 Aperture SFR fractions

Define

$$f_1 = \frac{\text{SubhaloSFRinHalfRad}}{\text{SubhaloSFR}}, \quad f_2 = \frac{\text{SubhaloSFRinRad}}{\text{SubhaloSFR}}. \quad (9)$$

Here f_1 is the fraction of total subhalo SFR inside $R_{\star,1/2}$, and f_2 is the fraction inside $2R_{\star,1/2}$. If $f_1 \simeq 1$, star formation is compact; if $f_1 \ll 1$, star formation is extended.

2.4 Convert SFR concentration to R_e

Use the same deprojected Sérsic form as Gao et al.:

$$\rho(r) = \rho_0 \left(\frac{r}{R_e} \right)^{-p} \exp \left[-b_n \left(\frac{r}{R_e} \right)^{1/n} \right], \quad (10)$$

with

$$b_n \simeq 2n - \frac{1}{3} + \frac{0.009876}{n}, \quad p \simeq 1 - \frac{0.6097}{n} + \frac{0.05563}{n^2}. \quad (11)$$

The 3D enclosed fraction is

$$F_n(< r | R_e) = \frac{\gamma \left[a_n, b_n (r/R_e)^{1/n} \right]}{\Gamma(a_n)}, \quad a_n = n(3 - p). \quad (12)$$

Solve

$$F_n(< R_{*,1/2} | R_e) = f_1, \quad F_n(< 2R_{*,1/2} | R_e) = f_2. \quad (13)$$

For one aperture, the inversion is

$$R_{e,1} = R_{*,1/2} \left(\frac{b_n}{P^{-1}(a_n, f_1)} \right)^n, \quad (14)$$

where P^{-1} is the inverse regularized lower incomplete gamma function. Similarly,

$$R_{e,2} = 2R_{*,1/2} \left(\frac{b_n}{P^{-1}(a_n, f_2)} \right)^n. \quad (15)$$

Use the logarithmic average as the default radius:

$$R_{e,\text{SFR}} = 10^{\frac{1}{2}[\log_{10} R_{e,1} + \log_{10} R_{e,2}]}. \quad (16)$$

Flag the subhalo if

$$\left| \log_{10} \left(\frac{R_{e,1}}{R_{e,2}} \right) \right| > 0.3, \quad (17)$$

because the SFR profile is probably too clumpy or disturbed to be represented by a smooth Sérsic profile.

2.5 Workflow

1. For each DMO halo/subhalo at a GC formation snapshot, identify the matched hydro subhalo.
2. Read `SubhaloSFR`, `SubhaloSFRinHalfRad`, `SubhaloSFRinRad`, and `SubhaloHalfmassRadType[:,4]`.
3. Convert $R_{*,1/2}$ from ckpc/h to physical kpc.
4. Compute f_1 and f_2 .
5. Choose a Sérsic index n ; use $n = 1.5$ as the fiducial and test $n = 1.0$ and $n = 2.0$.
6. Infer $R_{e,\text{SFR}}$ from the Sérsic CDF inversion.
7. Set $R_{e,\text{GC}} = f_{\text{GC}} R_{e,\text{SFR}}$, with f_{GC} usually set to 1 unless an explicit GC compactness bracket is desired.
8. Draw new GC radii from the same Sérsic cumulative profile used in the dynamical model.
9. If the match is missing, SFR is zero, the stellar radius is unresolved, or the two aperture estimates strongly disagree, fall back to Method 1.

2.6 Quality cuts

Use this method only when

$$\text{SubhaloSFR} > 0, \quad R_{\star,1/2} > 0, \quad 0 < f_1 \leq f_2 \leq 1. \quad (18)$$

A practical resolution cut is

$$\text{SubhaloLenType}[:,4] \gtrsim 100, \quad (19)$$

though stricter cuts are safer for low-mass high-redshift systems.

3 Connecting either R_e to GC spatial sampling

Both methods produce a replacement effective radius $R_{e,\text{GC}}$. The GC radius draw is then unchanged in structure:

1. Adopt a Sérsic index n and compute b_n , p , and $a_n = n(3 - p)$.
2. Draw $u \sim U(0, 1)$.
3. Solve

$$F_n(< r_{\text{GC}} \mid R_{e,\text{GC}}) = u. \quad (20)$$

Equivalently,

$$r_{\text{GC}} = R_{e,\text{GC}} \left[\frac{P^{-1}(a_n, u)}{b_n} \right]^n. \quad (21)$$

4. If the subsequent model only evolves galactocentric radius, store r_{GC} . If full 3D positions are needed, sample isotropic angles.

4 Recommended usage

Use Method 1 as the baseline because it is robust and works even when hydro catalogues are unavailable. Use Method 2 whenever hydro–DM matching and hydro group catalogues are available; it captures halo-to-halo variation in SFR compactness without requiring full gas-cell data. In production runs, keep the original Gao spin model as a control case, and compare the sunk IMBH/GC mass fractions under:

$$\{\text{spin model, empirical size–mass–redshift, SFR–concentration catalogue proxy}\}. \quad (22)$$

References

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